MetaService
Hybrid Compute Architecture

On-prem machines + AWS + Liquid Compute

Technical overview for Kurnia and Luca · 2026-11-10

> **The simple version:** MetaService manages the computers we own. AWS can run extra
> runtime.

Read this PDF

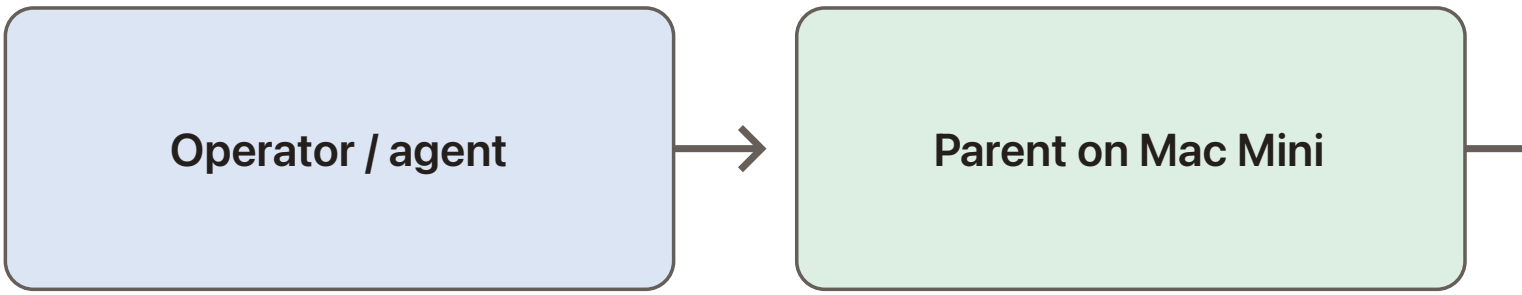
- **Page 2:** the parent, helpers, VMs, and existing services.
- **Page 3:** Compute, sandboxes, storage, and ownership.
- **Page 4:** the actual office cable/VLAN model.
- **Page 5:** why the gateway and ARP test matter.
- **Page 6:** the proposed hybrid AWS/Liquid Compute flow.
- **Page 7:** why ECS Anywhere is the first candidate.
- **Page 8:** phased rollout and rules for a safe proof of concept.

Architecture commit 03883a4. The AWS and Liquid Compute sections are proposals, not deployed featur

1. One parent, many machines

The **parent** coordinates inventory, placement, commands, usage, and audit. A native **hel**

MetaService request path



Layer	Responsibility
Parent on Mac Mini	Panel, API, machine inventory, placement, command IDs, retrie
macOS helper	Swift + Apple Container for managed VMs and external-servic
Linux helper	Rust + Incus for VMs/containers; discovers existing Docker/Inc
Runtime	Actually starts, stops, pauses, or deletes the workload.

> The parent decides **what** should happen. The helper performs it on the correct machi

2. Compute and ownership

- **Managed workloads:** created by MetaService and included in its placement budgets
- **External services:** already running on the machine; discovered without adoption.
- **Sandboxes:** API keys, expiration, pause/resume/kill, and usage metering.
- **Storage:** real disk maps and classification, separate from Compute.

Measurement	Meaning
Real machine memory	Actual used/cache/free, including external se
Managed allocation	Resources promised to managed workloads.
Placement capacity	What MetaService may still allocate after res

> **Important:** external services never enter managed placement accounting. Two contro

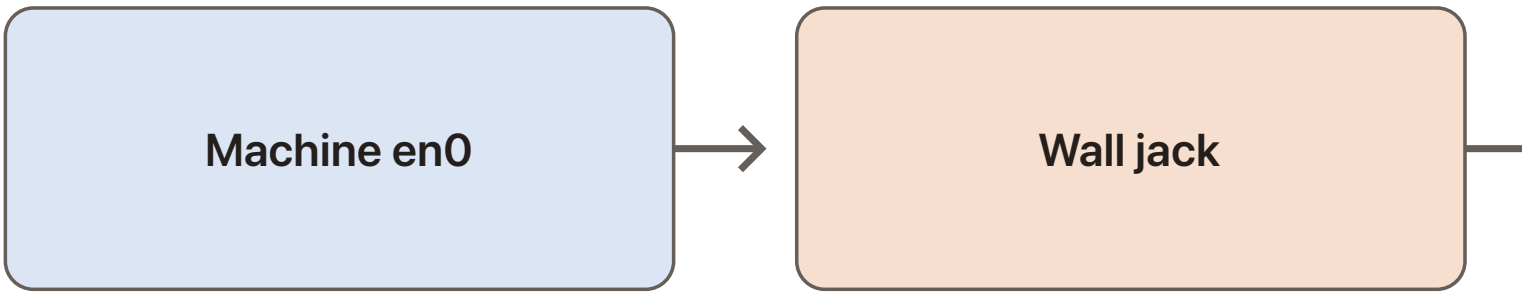
Current product boundaries

- External start/stop/restart requires explicit operator action and fresh target resolution
- External deletion is not offered.
- HTTP 202 means accepted, not finished; final command state comes from the helper
- Pause, resume, and kill exist. Fork/cloning and hosted billing are not shipped.

3. Office network: from cable to VLAN

The office network is a separate operational layer from MetaService. The wall-jack labels

Physical office connection



Wall jack	Switch port	Purpose
26192-O	ge-5/0/6	Primary public connection; alr
26191-O	ge-5/0/1	Secondary public connection f

Two networks

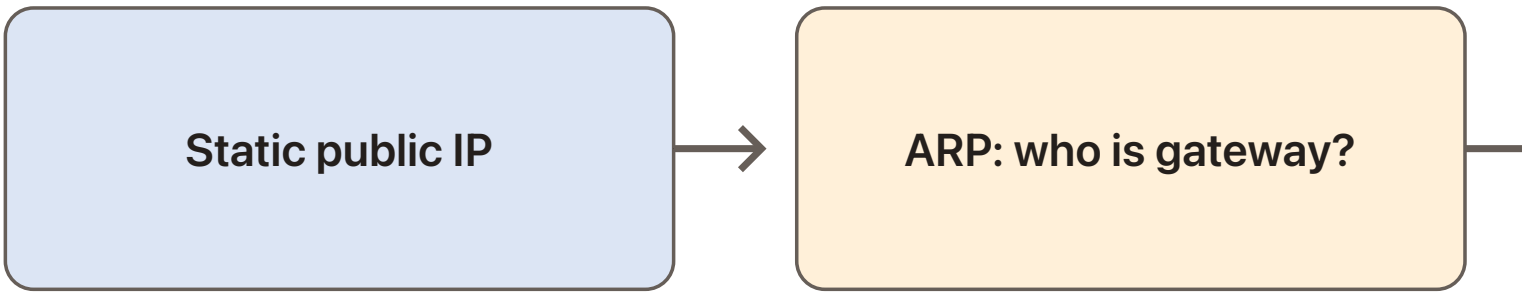
- **Member/floor VLAN:** DHCP supplies a private office address.
- **VLAN999:** static-only public connection; no DHCP address is supplied.

The secondary test machine uses Ethernet for the public connection and WiFi for its man

4. Why gateway + ARP prove the cable path

When the Mac has the correct public static configuration, it asks the local network: “Who

Layer-2 proof



> **Observed before WeWork enabled the port:** 100% gateway packet loss and no ARP mapping, VLAN, or MAC registration was responsible.

Why WeWork asks for a MAC address

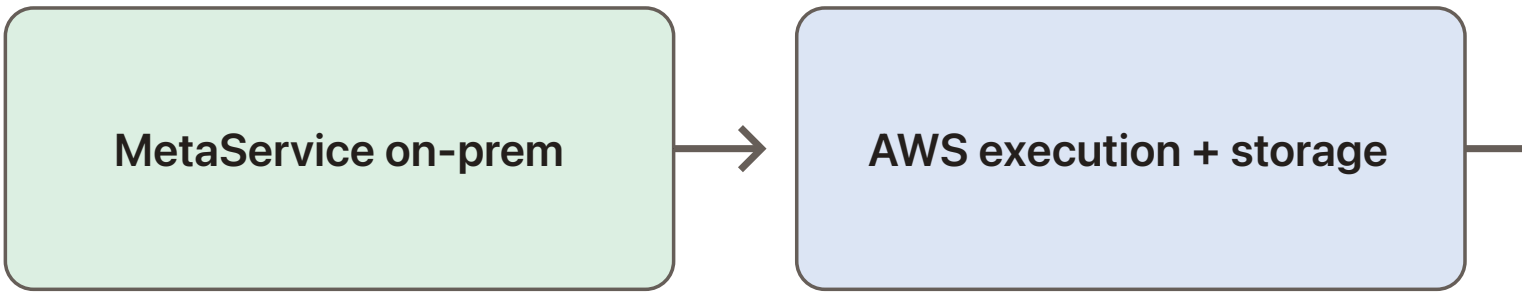
- Locate which switch port the machine actually reaches.
- Check whether that port carries the member VLAN or VLAN999.
- Apply port security or register the device if their policy requires it.

No local router/MikroTik is required when a machine holds the public IP directly. A router l firewall policy.

5. Proposed hybrid architecture

Execution, metering, and market settlement stay separate. Solid ownership boundaries p

Hybrid on-prem + cloud + market



System	Role
MetaService	On-prem Compute, ownership, VM/sandbox lifecycle, usage,
AWS	Cloud/container jobs, S3 results/checkpoints, queues, IAM, a
Liquid Compute	Market pricing, verification, matching, settlement, and physic

> **Liquid Compute is not a scheduler.** Its official API, authentication, supported hardwa

6. AWS choices

ECS Anywhere — first hybrid-container candidate

- One ECS control plane for compatible containers on AWS and on-prem.
- Uses the ECS agent, Systems Manager Agent, Docker, and external-instance registra
- Requires stable internet connectivity to AWS; it is not air-gapped.
- Existing tasks continue during control-plane disconnect, but scaling/mutations stop a
- GPU support still requires verification of our OS, CPU architecture, drivers, and devic

Direct EC2 adapter

Use this when MetaService needs to create/delete actual cloud VMs with strict quota and

Not the first choice: Outposts

Outposts deploys AWS-managed infrastructure at a site. It does not simply reuse our exis

> ECS Anywhere does not replace the MetaService macOS helper or VM lifecycle. AWS-
> external Docker services.

7. Safe rollout plan

Phase	Work
0. Discovery	Obtain official Liquid Compute API/participant documentation.
1. AWS read-only	Verify OS/CPU/GPU compatibility and estimate cost.
2. One test job	One reversible, non-sensitive container job using dedicated AWS resourc
3. Evidence export	Read-only verification/delivery bundle for Liquid Compute.

Non-negotiables

- No production GPU interruption.
- No migration of existing services.
- No office-LAN access from proof-of-concept workloads.
- No credentials in Git or documents.
- No automatic selling, settlement, or payout in the first integration.
- No shared workload ownership between controllers.

> The copy-ready PRD prompt is in `architecture/PROMPT-liquid-compute.md`. The detail